

30 Days SQL Micro Course Assignment

-- 1. Total Number of Patients

-- Write an SQL query to find the total number of patients across all hospitals.

```
SELECT SUM(patients_count) AS Total_Patients FROM hospital_data;
```

-- 2. Average Number of Doctors per Hospital

-- Retrieve the average count of doctors available in each hospital.

```
SELECT hospital_name, AVG(doctors_count) AS average_doctors_per_hospital
FROM hospital_data
GROUP BY hospital_name;
```

-- 3. Top 3 Departments with the Highest Number of Patients

-- Find the top 3 hospital departments that have the highest number of patients.

```
SELECT department, SUM(patients_count) AS Total_Patients
FROM hospital_data
GROUP BY department
ORDER BY total_patients DESC
LIMIT 3;
```

-- 4. Hospital with the Maximum Medical Expenses

-- Identify the hospital that recorded the highest medical expenses.

```
SELECT hospital_name, SUM(medical_expenses) AS Total_Expenses
FROM hospital_data
GROUP BY hospital_name
ORDER BY Total_Expenses DESC
LIMIT 1;
```

-- 5. Daily Average Medical Expenses

-- Calculate the average medical expenses per day for each hospital.

```
SELECT hospital_name,
       AVG(medical_expenses / (discharge_date - admission_date)) AS
Daily_Avg_Expense
FROM hospital_data
GROUP BY hospital_name;
```

-- 6. Longest Hospital Stay

-- Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

```
SELECT *, DATEDIFF(discharge_date, admission_date) AS stay_length
FROM hospital_data
ORDER BY stay_length DESC
LIMIT 1;
```

-- 7. Total Patients Treated Per City

-- Count the total number of patients treated in each city.

```
SELECT location AS city, SUM(patients_count) AS Total_Patients
FROM hospital_data
GROUP BY location;
```

-- 8. Average Length of Stay Per Department

-- Calculate the average number of days patients spend in each department.

```
SELECT department,
       AVG(discharge_date - admission_date) AS avg_stay_days
FROM hospital_data
GROUP BY department;
```

-- 9. Identify the Department with the Lowest Number of Patients

-- Find the department with the least number of patients.

```
SELECT department, SUM(patients_count) AS Total_Patients
FROM hospital_data
GROUP BY department
ORDER BY Total_Patients ASC
LIMIT 1;
```

- 10. Monthly Medical Expenses Report
- Group the data by month and calculate the total medical expenses for each month.

```
SELECT TO_CHAR(admission_date, 'YYYY-MM') AS month,  
       SUM(medical_expenses) AS total_expenses  
FROM hospital_data  
GROUP BY month  
ORDER BY month;
```